East West University
Department of Computer Science and Engineering

Course: CSE 475 (Machine Learning)
Section - 03

Submitted To :	**Submitted By :**
Dr. Md. Tauhid Bin Iqbal Assistant Professor Department of CSE East West University	**Name** : Hafsa Ferdousi **Student ID** : 2023-3-60-321 **Department** : CSE **Date** : 01 /07/ 2026

Lab Report:05

The clustering performance of **K-Means**, **K-Medoids**, and **Hierarchical Clustering** was evaluated using four different values of **K (2, 3, 4, 5, and 6)**. The quality of clustering was measured using three evaluation metrics:

- **WCSS (Within-Cluster Sum of Squares):** Lower values indicate more compact clusters.
- **Silhouette Score:** Higher values indicate better cluster separation and cohesion.
- **Davies-Bouldin Index (DBI):** Lower values indicate better clustering performance.

The results show that increasing the number of clusters generally improves clustering quality across all three algorithms. WCSS decreases consistently as K increases because data points become more compact within clusters. Similarly, the DBI also decreases with higher K values, indicating better separation between clusters. The Silhouette Score shows slight improvement as K increases, although the improvement is not always consistent for every algorithm.

K-Means Clustering Analysis

```
K=2 | WCSS=399.770 | Silhouette=0.1799 | DBI=2.0594
K=3 | WCSS=349.714 | Silhouette=0.1601 | DBI=1.8064
K=4 | WCSS=309.930 | Silhouette=0.1699 | DBI=1.6182
K=5 | WCSS=272.216 | Silhouette=0.1881 | DBI=1.4106
K=6 | WCSS=243.131 | Silhouette=0.1954 | DBI=1.3704
```

Among all tested values, **K = 6** produced the best performance for K-Means with:

- **WCSS = 243.131**
- **Silhouette Score = 0.1954 (Highest)**
- **DBI = 1.3704 (Lowest)**

These results indicate that K-Means achieved the most compact and well-separated clusters when six clusters were used. Compared to the other algorithms, K-Means consistently produced higher Silhouette Scores and lower DBI values, demonstrating superior clustering performance.

K-Medoids Clustering Analysis

```
K=2 | Cost=194.639 | WCSS=432.178 | Silhouette=0.1469 | DBI=2.0305
K=3 | Cost=189.075 | WCSS=424.336 | Silhouette=0.1132 | DBI=2.0182
K=4 | Cost=172.731 | WCSS=357.441 | Silhouette=0.1432 | DBI=1.7522
K=5 | Cost=172.706 | WCSS=350.683 | Silhouette=0.1170 | DBI=1.6557
K=6 | Cost=159.096 | WCSS=321.695 | Silhouette=0.1145 | DBI=1.6183
```

K-Medoids also showed a reduction in WCSS and DBI as K increased. However, its Silhouette Scores remained relatively low throughout the experiment.

The best performance was observed at **K = 6**, where:

- **WCSS = 321.695**
- **Silhouette Score = 0.1145**
- **DBI = 1.6183**

Although clustering quality improved with increasing K, K-Medoids performed worse than K-Means and Hierarchical Clustering according to both Silhouette Score and DBI. This suggests that K-Medoids were less effective for this dataset.

Hierarchical Clustering

```
K=2 | WCSS=417.946 | Silhouette=0.1537 | DBI=2.1947
K=3 | WCSS=365.852 | Silhouette=0.1518 | DBI=1.9144
K=4 | WCSS=325.863 | Silhouette=0.1602 | DBI=1.7100
K=5 | WCSS=290.965 | Silhouette=0.1553 | DBI=1.6429
K=6 | WCSS=258.881 | Silhouette=0.1800 | DBI=1.4776
```

Hierarchical Clustering also exhibited gradual improvement as the number of clusters increased.

The best result was obtained at **K = 6**, with:

- **WCSS = 258.881**
- **Silhouette Score = 0.1800**
- **DBI = 1.4776**

Its performance was better than K-Medoids but slightly inferior to K-Means. The higher Silhouette Score and lower DBI compared to K-Medoids indicate that Hierarchical Clustering generated more meaningful clusters.

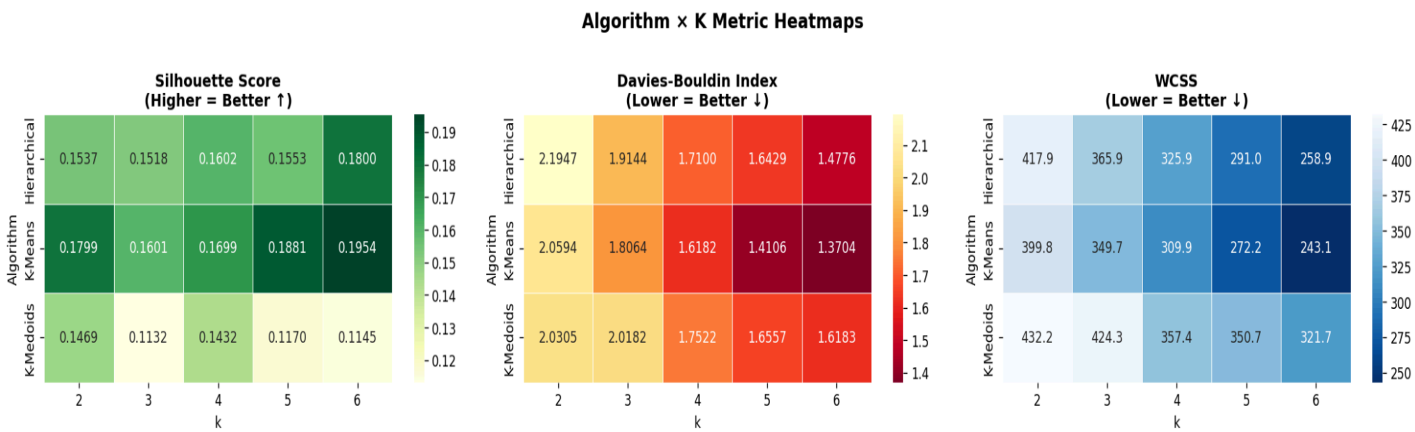
Overall Analysis

	Algorithm	k	WCSS	Silhouette	DBI
0	K-Means	2	399.770000	0.179900	2.059400
1	K-Medoids	2	432.178000	0.146900	2.030500
2	Hierarchical	2	417.946000	0.153700	2.194700
3	K-Means	3	349.714000	0.160100	1.806400
4	K-Medoids	3	424.336000	0.113200	2.018200
5	Hierarchical	3	365.852000	0.151800	1.914400
6	K-Means	4	309.930000	0.169900	1.618200
7	K-Medoids	4	357.441000	0.143200	1.752200
8	Hierarchical	4	325.863000	0.160200	1.710000
9	K-Means	5	272.216000	0.188100	1.410600
10	K-Medoids	5	350.683000	0.117000	1.655700
11	Hierarchical	5	290.965000	0.155300	1.642900
12	K-Means	6	**243.131000**	**0.195400**	**1.370400**
13	K-Medoids	6	321.695000	0.114500	1.618300
14	Hierarchical	6	258.881000	0.180000	1.477600

Based on the experimental results, **K = 6** is the optimal number of clusters for all three clustering algorithms. Among them, **K-Means** is the most suitable algorithm for this dataset because it achieved:

- The **lowest WCSS (243.131)**, indicating the most compact clusters.
- The **highest Silhouette Score (0.1954)**, indicating the best cluster separation.
- The **lowest Davies-Bouldin Index (1.3704)**, indicating the highest clustering quality.

Therefore, **K-Means with K = 6** is selected as the optimal clustering model for this dataset, as it provides the best balance of cluster compactness, separation, and overall clustering quality



The heatmaps and comparison table further confirm that **K-Means with K = 6** achieved the best overall clustering performance. It has the darkest color in the Silhouette heatmap (highest score) and the lightest color in the WCSS and DBI heatmaps (lowest values), indicating compact and well-separated clusters. Hierarchical Clustering ranked second, while K-Medoids showed the weakest performance across the evaluation metrics.

Code Link: Clustering.ipynb